**DE–8530** **11**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization – Plant Biotechnology)
DEGREE EXAMINATION, MAY 2014.

PLANT DIVERSITY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Describe the life cycle of Rhodophyceae. (7)

   (b) Give detailed account on fungal classification based on Alexopoulos and Mims (1979). (7)

   (c) Briefly mentioned about general characteristic features of lichens. (6)

2. (a) Describe the distinguishing features of Bryophytes. (6)

   (b) Describe the life cycle of *Funaria* with the help of labeled Sketches. (7)

   (c) Give a comparative account of the gametophytes of *Lycopodium* and *Sphenophyllum*. (7)

3. (a) Write an essay on the thallus organization in Chlorophycean members. (7)

   (b) Write in detailed account on general characters of Deuteromycetes. (6)

   (c) Write an essay on life cycle of *Gnetum*. (7)

4. (a) Give an account of the sporophyte of *Polytrichum*. (7)

   (b) Write briefly about fossil Pteridophyte with suitable examples. (6)

   (c) Write an essay on Smith classification of algae. (7)

5. (a) Give a comparative account between *Marchantia* and *Anthoceros*. (7)

   (b) Explain the life history of *Lycopodium*. (7)

   (c) Explain the general characters of Gymnosperms. (6)

6. (a) Describe reproduction methods in Cyanophyceae. (6)

   (b) Write an essay on Mariculture. (7)

   (c) Describe the methods of reproduction in Lichens. (7)

7. (a) Give a brief account on sporophyte of *Marchantia*. (6)

   (b) Write an essay on classification of Pteridophytes based on Reimers. (7)

   (c) Give a detailed account on fossil Gymnosperms with suitable example. (7)

8. (a) Give a detailed account on reproduction of various fungi. (7)

   (b) Write a short account on internal struckere of lichen. (6)

   (c) Discuss the structural variation in the gametophytes of *Sphagnales* and *Andreales*. (7)

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**DE–8531** **12**

DISTANCE EDUCATION

M.Sc. (Botany) DEGREE EXAMINATION, MAY 2014.

(Specialization-Plant Biotechnology)

PLANT TAXONOMY

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each questions carries 20 marks.

Draw diagram wherever necessary

($5 \times 20 = 100$)

1. (a) Define species, biotype, ecad, and ecoype. (8)

   (b) Explain binomial system of nomenclature with example. (6)

   (c) Narrate various theories in biological classification. (6)

2. (a) Give an account on molecular systematics. (6)

   (b) Write examples for structural systematics. (6)

   (c) Discuss background of plant classification briefly. (8)

3. (a) Explain Bentham and Hooker system of classification. (10)

   (b) Write an account on the merits and demerits of Takhtajan classification. (6)

   (c) Give an account of the medicinally important plants in Boraginaceae. (4)

4. (a) Narrate Hutchinson system of classification. (10)

   (b) Discuss merits and demerits of Engler and Prantl system of classification. (6)

   (c) Write about the economically important plants in Rhamnaceae. (4)

5. (a) Write an account on biosystematics. (8)

   (b) Comment on chemotaxonomy and its application. (6)

   (c) Discuss the relevance of numerical taxonomy. (6)

6. (a) Give an account on plant geography. (6)

   (b) Define floristics and cite examples. (8)

   (c) Write an account on modern multi-disciplinary approaches to Taxonomy. (6)

7. (a) Expand ICBN and comment its role. (5)

   (b) Define principle of priority, valid name and correct name with examples. (9)

   (c) What is type? Give its types with examples. (6)

8. (a) Write an account on the family of Orchidaceae. (7)

   (b) Ranunculaceae is a primitive family – substantiate by diagnostic features. (7)

   (c) Explain the diagnostic features of Loranthaceae. (6)

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**DE–8532** **13**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization – Plant Biotechnology) DEGREE EXAMINATION, MAY 2014.

ANATOMY AND EMBRYOLOGY

(2008 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each questions carry 20 marks.

($5 \times 20 = 100$)

1. (a) Write note on Tunica-Corpus theory. (6)

   (b) Give an account on phylogenetic specialization of phloem. (10)

   (c) Write notes on cambium. (4)

2. (a) Give an account on light microscopic structure of the cell wall. (10)

   (b) Write an essay on seasonal activity of cambium. (5)

   (c) What are the various theories of apical meristem. (5)

3. (a) Explain the primary structure of root. (8)

   (b) Explain the molecular aspects of developing vegetative organs. (6)

   (c) Describe the anatomical structure of dicot leaf. (6)

4. (a) How will you identify the wood? Explain with suitable examples. (10)

   (b) Explain the natural defects of wood. (5)

   (c) What are the commercial woods of South India? (5)

5. (a) Write a short note on mechanical properties of wood. (6)

   (b) Write short notes on knots of wood. (6)

   (c) Give an account on anther development. (8)

6. (a) Give an illustrated account on megasporogenesis. (10)

   (b) Explain the types of endosperm. (5)

   (c) Comment on apomixis. (5)

7. (a) Explain the pollen-stigma compatibility and how will you overcome this. (10)

   (b) Enumerate the different types of vegetative reproduction in plants. (6)

   (c) Describe the process of microsporogenesis. (4)

8. (a) Write the role of polyembryony in plant improvement programmes. (10)

   (b) Briefly explain the molecular aspects of higher plant reproduction. (6)

   (c) Write short note on apospory. (4)

**DE–8533** **14**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization-Plant Biotechnology) DEGREE EXAMINATION, MAY 2014.

PLANT TISSUE CULTURE

(2008 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) How are callus cultures maintained? (5)

   (b) Write notes on major types of media. (5)

   (c) Write an essay on suspension cultures. (10)

2. (a) State the applications of micropropagation. (5)

   (b) Explain biotransformation. (5)

   (c) Explain the methods to eliminate viruses in plants. (10)

3. (a) Write short notes on organogenesis. (5)

   (b) Briefly describe the procedure involved in synthetic seeds production. (5)

   (c) Write a detailed account on somaclonal variations. (10)

4. (a) Write an account on meristem culture. (5)

   (b) Write a brief account on flower culture. (5)

   (c) Write an essay on androgenesis. (10)

5. (a) Explain the enzymatic methods of protoplast isolation. (5)

   (b) Give a brief notes on in vitro production of secondary metabolites. (5)

   (c) Give a detailed account on importance and achievements of protoplast research. (10)

6. (a) Briefly describe the basic principles of plant tissue culture. (5)

   (b) Write notes on embryogenesis. (5)

   (c) State any five secondary metabolites obtained from plant cell cultures and their applications. (10)

7. (a) Write notes on fusigenic agents in tissue culture experiments. (5)

   (b) Enumerate the importance of hormones in plant regeneration. (5)

   (c) Write an essay on major types of suspension cultures. (10)

8. (a) Discuss about various factors regulating organogenesis and embryogenesis. (5)

   (b) Explain the various factors known to influence callus culture. (5)

   (c) Describe the procedure involved in the *invitro* fusion of isolated protoplasts to form a hybrid plant. (10)

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**DE–8534** **15**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization – Plant Biotechnology) DEGREE EXAMINATION, MAY 2014.

PLANT MOLECULAR BIOLOGY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Describe the targeting of proteins to chloroplast compartments. (6)

   (b) Write short notes on chloroplast-enclosed gene. (6)

   (c) Explain chloroplast genome organization. (8)

2. (a) Write the classification and function of seed storage protein. (7)

   (b) Write an essay on gene regulation in eukaryotes. (8)

   (c) List out the uses of transposons. (5)

3. (a) Give detailed account on symbiotic nitrogen fixation in legumes. (8)

   (b) Comment on co integrate and binary vectors. (5)

   (c) Outline the method of T-DNA transfer. (7)

4. (a) Explain any two physical methods of direct transformation. (7)

   (b) Comment on drought tolerant plants. (8)

   (c) Describe silicon carbide WHISKER. (5)

5. Write short notes on :

   (a) Tagging of plant gene. (5)

   (b) Gene mapping. (5)

   (c) Write an essay on bioremediation of contaminated soil and waste land. (10)

6. (a) Write an account of eukaryotic genome organization. (8)

   (b) Describe the tumour inducing principle and genetic organization of Ti plasmid. (7)

   (c) Write an account of plasmid as vectors. (5)

7. (a) Analyse the sequential stages of gene cloning techniques. (6)

   (b) Discuss - microbes in bioleaching. (8)

   (c) Give an account of bioremediation of hydrocarbons. (6)

8. (a) Give an account of development and application of transgenic plants. (8)

   (b) Write short notes on production of bioplastics. (7)

   (c) Discuss the cytoplasmic male sterility. (5)

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**DE–8535** **21**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization – Plant Biotechnology) DEGREE EXAMINATION, MAY 2014.

PLANT PHYSIOLOGY AND BIOCHEMISTRY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) What is transpiration? What are the factors affecting transpiration? (5)

   (b) Explain water absorption and transport in plant. (4)

   (c) Write short notes on:

   (i) Osmosis (3)

   (ii) Active transport (4)

   (iii) Passive transport. (4)

2. (a) Discuss briefly about Hatch and Slack pathway. (8)

   (b) Explain : Photosynthesis in chloroplast. (8)

   (c) Write short notes on: water potential. (4)

3. (a) Discuss briefly about electron transport in mitochondria. (6)

   (b) Explain : Preparatory phase and payoff phase of EMP. (6)

   (c) How much of ATP molecule synthesised in Glycolysis and Krebs cycle? Describe its the stages. (8)

4. (a) What is stereoisomers? Explain with examples. (6)

   (b) Discuss about amino acid biosynthesis. (8)

   (c) Write about the characteristics and classification of proteins with examples. (6)

5. (a) Michaelis-Menton equation. (8)

   (b) Short notes on: Biosynthesis of cholesterol. (6)

   (c) Short notes on: Peptide bond and polypeptide chain. (6)

6. (a) Explain about the enzyme properties, classifications and nomenclature. (8)

   (b) Write about purine synthesis. (6)

   (c) Describe the structure and composition of DNA. (6)

7. (a) Discuss: Physical, chemical properties and biological important of amino acids. (10)

   (b) What are the factors influencing the enzyme action? (5)

   (c) Short notes on: Fatty acids. (5)

8. (a) Describe about the biological nitrogen fixation. (10)

   (b) Short notes on: lipids. (5)

   (c) Describe the structure of proteins. (5)

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**DE–8536** **22**

DISTANCE EDUCATION

M.Sc. (Botany – Specialization Plant Biotechnology) DEGREE EXAMINATION, MAY 2014.

CELL BIOLOGY, GENETICS AND PLANT BREEDING

(2008 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) Describe the structure and function of Mitochondria. (7)

   (b) Enumerate the functions of the endoplasmic reticulum. (6)

   (c) Enumerate the functions of the nucleus. (7)

2. (a) What are the postulates of the fluid mosaic model? (7)

   (b) Write short notes on the trafficking of proteins from the endoplasmic reticulum to the Golgi complex. (6)

   (c) Describe the events taking place during mitosis. (7)

3. (a) Describe the structure and function of the centrosome. (6)

   (b) List out the functions of the plasma membrane. (7)

   (c) Enumerate and describe the various types of lysosomes. (7)

4. (a) Write short notes on modifiers, suppressors and pleiotropic genes. (6)

   (b) List the applications of Hardy–Weinberg's law in calculating gene frequencies in a population. (7)

   (c) Enumerate the different type of mutagenic agents, adding a note on each type. (7)

5. (a) Write short notes on segregation, distortion and selfish genes. (6)

   (b) At what stage of meiosis does crossing over occur? Describe one of the classical experiments in which it was detected cytologically. (8)

   (c) Enumerate the types of male sterility in plants, and briefly describe each type. (6)

6. (a) Describe the methods by which apomictic plants can be breed. (6)

   (b) List the objectives of plant breeding. (6)

   (c) Write a note on any one method employed for the breeding of self pollinated plants. (8)

7. (a) What is a point mutation? Enumerate the types of point mutation. (6)

   (b) Give an account of inbreeding depression theories. (7)

   (c) What are polyploids, add a note on its significance in plant breeding? (7)

8. (a) What are Morgan's views on linkage? (7)

   (b) Write a short note on the process of synapsis, which occurs during meiosis. (7)

   (c) Explain test cross. (6)

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**DE–8537** **23**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization Plant Biotechnology) DEGREE EXAMINATION, MAY 2014.

BIOTECHNIQUES IN BOTANY

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) Write short notes on mechanism of polymerization in PAGE. (8)

   (b) How will you determine the molecular weight of the DNA on agarose gel electrophoresis? (7)

   (c) Write the principle of isoelectirc focusing. (5)

2. (a) Write about the different types of centrifuges. (7)

   (b) Principle of confocal microscopy. (6)

   (c) Write the procedure for sample preparation of TEM. (7)

3. (a) List out the different applications of radioactive isotopes. (8)

   (b) Write the principle of liquid scintillation spectrometry. (7)

   (c) Write a note on autoradiography. (5)

4. (a) What are the different types of blotting techniques? (8)

   (b) Write detailed notes on GCMS. (6)

   (c) Write the various applications of PCR. (6)

5. (a) Write about the various types of DNA sequencing. (7)

(b) How DNA fingerprinting is being applied in the field of botany? (6)

(c) How will you compare the evolutionary relationship between the organisms using molecular techniques? (7)

6. (a) Principle of MALDI–TOF. (7)

(b) Write short notes on principle and applications of RFLP. (7)

(c) Working mechanism of HPLC. (6)

7. (a) Comment on the relationship between genome, transcriptome and proteomes of plants. (8)

(b) Write the applications of microarray. (6)

(c) How will you find out evolutionary relationship of an organism using a bioinformatics tool? (6)

8. (a) Write short notes on protein data base. (6)

(b) Principle of phase contrast microscopy. (7)

(c) Explain about RAPD. (7)

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**DE–8538** **24**

DISTANCE EDUCATION

M.Sc. DEGREE EXAMINATION, MAY 2014.

Botany (Specialization Plant Biotechnology)

BIODIVERSITY CONSERVATION AND IPR

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) Give an account of genetic diversity. (5)

   (b) Write critical notes on mega diversity centres. (5)

   (c) Discuss in detail about values and uses of biodiversity. (10)

2. (a) Comment on endemism. (6)

   (b) Distinguish between extinct plants and endangered plants. (7)

   (c) Explain hot spot diversity. Add notes on criterias for recognising hot spots. (7)

3. (a) Write a brief account of ex situ conservation. (6)

   (b) Write short notes on buffer zones of protected areas. (7)

   (c) Enumerate the significance of in situ conservation. (7)

4. (a) Write a brief account of world trade organization. (5)

   (b) Describe the different types of intellectual property rights. (8)

   (c) What are the conditions for granting protection for new plant varieties. (7)

5. (a) Give an account of threatened categories. (10)

   (b) Describe the various forest communities of India. (5)

   (c) Write a brief note on wild life sanctuaries. (5)

6. (a) Write short note on Savannah. (7)

   (b) Discuss in detail about phytogeographic zones of India. (7)

   (c) Comment on centres of diversity. (6)

7. (a) Distinguish between process patents and product patent. (5)

   (b) What are the various steps involved in patenting biological materials. (10)

   (c) Describe the patent case of turmeric. (5)

8. (a) Discuss in detail about recombinant DNA safety guidelines. (6)

   (b) What is biosafety? Point out the good laboratory practices for a biological lab. (7)

   (c) Write a critical commentary on "Ricetech". (7)

**DE–8539** **25**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization plant Biotechnology) DEGREE EXAMINATION, MAY 2014

PLANT GENETIC ENGINEERING

(2008 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Write an essay on endonucleases and exonucleases. (8)

   (b) Add a note on Maxam and Gillbert technique. (5)

   (c) Write short notes on scope, importance and application of plant genetic engineering. (7)

2. (a) What are reporter genes? Describe their role in selection of recombinants. (8)

   (b) Explain the role of selectable markers in screening methods. (7)

   (c) Discuss about antibiotic sensitivity assay. (5)

3. (a) Describe any three methods of gene transfer. Add a note on their advantages and disadvantages. (8)

   (b) Explain the calcium phosphate mediated DNA transfer. (6)

   (c) Discuss about AFLP. (6)

4. (a) What is antisense RNA? Explain its activity in genetic engineering? (7)

   (b) Write short notes on proteinase inhibitor and add a note on their advantages and limitation. (8)

   (c) Comment on cross protection satellite RNA. (5)

5. (a) Write an essay on any two molecular markers. (6)

   (b) Discuss about transplastomic plants. (8)

   (c) Add a note on Microsatellites. (6)

6. (a) Write an essay on the enzymes involved in genetic engineering. (10)

   (b) Discuss the role of Ti-plasmid as vectors to introduce foreign genes. (5)

   (c) Explain the method of chloroplast transformation method and its utility. (5)

7. (a) Write the role of coat protein in virus resistant plants. (6)

   (b) Describe PR proteins. (8)

   (c) Write short account of strategies to provide glyphosphate resistance to plants. (6)

8. (a) Write an essay on genetic engineering for male sterility. (6)

   (b) What are the strategies to develop water deficit tolerant plants? (6)

   (c) Describe the plant genome project. (8)

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